

MASTER_SECRET (server)
+ application, user,
modality, key version v

HKDF-SHA256
→ seeds S_P, S_T, S_π

key v

key v

Enrollment

Unit embedding e
(face / voice centroid)

Keyed transform:
 $W_v e$ (orthonormal rows)
→ $1[\cdot > t_v] \rightarrow \pi_v$

256-bit protected
template

Authentication

Fresh embedding e'
(one sample)

Same transform,
same key
(ACTIVE set's version v)

Hamming similarity h
→ calibrated metric
→ threshold

Store: 4 template sets, key versions $v \dots v+3$ (1 ACTIVE, 3 STANDBY)

Hand gesture: each weighted 13-D frame $\times$ keyed orthonormal Q_v ;
distances (and DTW) are preserved exactly.
Revocation: new key version $\rightarrow$ unrelated template.
Key must remain protected: BioHash weakens and Q_v is invertible if it leaks.

Solid: enrollment. Dashed: authentication.

No formal non-invertibility proof is claimed for BioHash.